

ID m° 1.

Ex 1. Rocq

Ex 2. Rocq

Ex 3. IF $\Gamma \vdash A$ is provable in NJ, take some proof tree in NJ for it, it is a valid proof tree in NJ + R, thus $\Gamma \vdash A$ is provable in NJ + R.

Conversely, take a proof tree π of $\Gamma \vdash A$ in NJ + R, by admissibility of R, every occurrence of R in π can be replaced by part of a tree that doesn't use R. Doing this starting from the leaves yield a valid proof tree for $\Gamma \vdash A$ in NJ.

Ex 4.

1. Take a proof tree of $\Gamma, A, B \vdash C$. Replace each occurrence of

$$\frac{}{\Gamma, A, B, \Delta \vdash A} \text{Ax}$$

$$\frac{\frac{}{\Gamma, A \wedge B, \Delta \vdash A \wedge B} \text{Ax}}{\Gamma, A \wedge B, \Delta \vdash A} \wedge \text{eg}$$

and

$$\frac{}{\Gamma, A, B, \Delta \vdash B} \text{Ax}$$

with

$$\frac{\frac{}{\tilde{\Delta} \vdash A \wedge B} \text{Ax}}{\tilde{\Delta} \vdash B} \wedge \text{ed},$$

also replacing every sequent $\Gamma, A, B, \Delta \vdash D$ with $\Gamma, A \wedge B, \Delta \vdash D$.

2. By induction on the size of Γ , we apply Q1.

3. No, derivable rules cannot remove hypotheses (with the rules in NJ).

Ex 5.

$$\begin{array}{l}
 1. \quad \frac{\Xi, \Gamma \vdash A}{\Xi, \Delta \vdash B} \text{ admissible iff } (\Xi, \Gamma \vdash A \text{ implies } \Xi, \Delta \vdash B) \\
 \text{iff } (\Xi \vdash \bigwedge \Gamma \Rightarrow A \text{ implies } \Xi \vdash \bigwedge \Delta \Rightarrow B) \quad \downarrow \text{Ex h.} \\
 \text{iff } \Xi \vdash (\bigwedge \Gamma \Rightarrow A) \Rightarrow (\bigwedge \Delta \Rightarrow B) \\
 \text{iff } \Xi \vdash (\bigwedge \Gamma \Rightarrow A \wedge \bigwedge \Delta \Rightarrow B) \quad \downarrow \text{currying}
 \end{array}$$

2. Similarly to Q1, we know

$$\Xi, \Gamma \vdash (\bigwedge \Delta \Rightarrow A) \Rightarrow B$$

is provable, thus, by repeated currying, writing $\Delta = F_1, \dots, F_n$, the sequent

$$\Xi, \Gamma \vdash (F_1 \Rightarrow \dots \Rightarrow F_n \Rightarrow A) \Rightarrow B$$

is provable; take π a proof tree for it.

$$\begin{array}{c}
 \frac{\frac{\pi}{\Xi, \Gamma \vdash \Phi \Rightarrow B}}{\Xi, \Gamma \vdash B} \quad \frac{\frac{\text{assumption}}{\Xi, \Gamma, \Delta \vdash A}}{\Xi, \Gamma \vdash F_1 \Rightarrow \dots \Rightarrow F_n \Rightarrow A} \xrightarrow[\Rightarrow_E]{\text{repeated } \Rightarrow_I}
 \end{array}$$

Ex 6.

1.

$$\frac{\frac{\frac{\pi_1}{\Gamma \vdash A}}{\Gamma, \Delta \vdash A} \text{ wk} \quad \frac{\frac{\pi_2}{\Delta \vdash B}}{\Gamma, \Delta \vdash B} \text{ wk}}{\Gamma, \Delta \vdash A \wedge B} \text{ \(\wedge I\)}$$

2.

$$\frac{\frac{\Gamma \vdash \neg A}{\Gamma, \Delta \vdash \neg A} \text{ wk} \quad \frac{\Delta \vdash A}{\Gamma, \Delta \vdash A} \text{ wk}}{\Gamma, \Delta \vdash \perp} \neg E$$

$$\frac{\frac{\Gamma \vdash A \vee B}{\Gamma, \Delta, \Sigma \vdash A \vee B} \text{ wk} \quad \frac{\frac{\Delta \vdash A \vdash C}{\Gamma, \Delta, \Sigma, A \vdash C} \text{ wk} \quad \frac{\Sigma \vdash B \vdash C}{\Gamma, \Delta, \Sigma, B \vdash C} \text{ wk}}{\Gamma, \Delta, \Sigma \vdash C} \vee E$$

$$\frac{\frac{\Gamma \vdash A \Rightarrow B}{\Gamma, \Sigma \vdash A \Rightarrow B} \text{wb}}{\Gamma, \Sigma \vdash B} \quad \frac{\frac{\Delta \vdash B}{\Gamma, \Delta \vdash B} \text{wR}}{\Rightarrow E.}$$

Ex 7.

Define $C_T := \perp \Rightarrow \perp$ and $C_{\neg A} := A \Rightarrow \perp$.

$$\frac{\frac{\frac{}{Ax}}{\Gamma, \perp \vdash \perp} \Rightarrow I}{\Gamma \vdash C_T} \text{"TI"}$$

$$\frac{\Gamma \vdash C_{\neg A} \quad \Gamma \vdash A}{\Gamma \vdash \perp} \Rightarrow E. \quad \neg E.$$

$$\frac{\Gamma, A \vdash \perp}{\Gamma \vdash C_{\neg A}} \Rightarrow I \quad \neg I$$

Ex 8. Rocq

Ex 9. Rocq

$\neg_R A$ is equivalent to $\neg A$ iff $\vdash \neg R$ is provable.

10m° 2.

I) Preliminaries

Exercise 1.

Take R a relation over λ -terms.

Then for all $M \rightarrow_R N$, for all $C[\]$, we have $C[M] \rightarrow_R C[N]$

by def. of $\rightarrow_R$.

Exercise 2

By induction on $M R^* N$,

if $M R^0 N$, i.e. $M = N$, then $C[M] R^0 C[M] = C[N]$.

if $M R N R^* P$ then by ih $C[N] R^* C[P]$

and $C[M] R C[N]$ by congruence.

II) the complexity of the simply-typed β -reduction.

Exercise 3.

For every $m \in \mathbb{N}$,

$$D_{A_m} = \lambda f_{m+2} : A_{m+2} \rightarrow A_m. \lambda f_m : A_m, \underbrace{f_{m+1} (f_{m+1} f_m)}_{: A_m} : \underbrace{A_{m+1} \rightarrow A_{m+1}}_{= A_{m+2}}$$

By induction on $k \in \mathbb{N}$, we show

$$T_{A_m, k} : A_{m+2} \rightarrow A_{m+3} \rightarrow \dots \rightarrow A_{m+2+k}$$

$$T_{A_m, 0} = D_{A_m} : A_{m+2}$$

$$T_{A_m, k+1} = T_{A_{m+1}, k} \underbrace{D_{A_m}}_{A_{m+2}}$$

Exercice 4.

By induction on k ,

- $T_{A_n, 0} = \mathcal{D}_{A_n}$ is already in β -normal form

$$2f_{m+1} \cdot 2f_m \cdot f_{m+1}^2 \cdot f_m$$

as $\sigma(1) = 2$

$$\bullet T_{A_n, k+1} = T_{A_{n+1}, k} D_{A_n} \xrightarrow[\beta]{\circ^* \left(\lambda_{f_{n+2}} \lambda_{f_{n+2}} \lambda_{f_{n+2}} \lambda_{f_{n+2}}^{2(k+1)} \right)} \left(\lambda_{f_{n+1}} \lambda_{f_n} f_{n+1}^2 f_n \right)$$

$$\rightarrow \rho^{\lambda f_{m+1}} \left(\lambda_{f_{m+1}} \cdot \lambda_{f_m} \cdot \frac{f_{m+1}^2}{f_m(f_m+1)} \right)^{\overline{O(kes)}} f_{m+1}$$

$$-\frac{\partial}{\partial \beta} \lambda f_{m+1} \cdot \lambda f_m \cdot f_{m+1}^{2^{(k+1)}} f_m$$

II | Taming β -reduction using the size of types.

Exercise 5. By induction on M .

$$\begin{array}{ccccc} x \xrightarrow{\circ_{\beta}^*} x & (x, M) \xrightarrow{\circ_{\beta}^*} \lambda x. M^* & MN \xrightarrow{\circ_{\beta}^*} M^* N & & \\ & \text{"} & & & \\ & (\lambda x. M)^* & \xrightarrow{\circ_{\beta}^*} M^* N^* & & \end{array}$$

$$\begin{aligned} (\lambda x. L) N &\longrightarrow_{\beta}^* (\lambda x. L^*) N \\ &\longrightarrow_{\beta}^* (\lambda x. L^*) N^* \\ &\longrightarrow_{\beta}^* L^* \{N^*/x\}. \end{aligned}$$

Exercise 6.

$$\begin{aligned} (\lambda x. x y) (\lambda a. a)^* &= (x y) \{ \lambda a. a / x \} \\ &= (\lambda a. a) y \longrightarrow_{\beta} y \end{aligned}$$

so not a β -normal form.

Some of the β -redexes in M are still present in M^* .

Exercise 7.

$$\begin{aligned}
 \omega_m(M \{N/x\}) &= \max \left\{ \omega(R) \mid R \text{ is a } \beta\text{-reduct in } M \{N/x\} \right\} \\
 &= \max \left(\left\{ \omega(R) \mid R \text{ is a } \beta\text{-reduct in } M \right\} \right. \\
 &\quad \left. \cup \left\{ \omega(R) \mid R \text{ is a } \beta\text{-reduct in } N \right\} \right. \\
 &\quad \left. \cup \left\{ \omega((\lambda x.u) a) \mid (\lambda x.u) = N, x a \in M \right\} \right) \\
 &= \max(\omega_m(M), \omega_m(N), \text{size}(A)).
 \end{aligned}$$

We can prove this by induction on M :

- $\omega_m(x \{N/x\}) = \omega_m(N) \leq \max(\dots)$
- $\omega_m((\lambda y.M) \{N/x\}) = \omega_m(\lambda y.(M \{N/x\})) = \omega_m(M \{N/x\}) \leq \max(\dots)$
- $\omega_m((x M) \{N/x\}) = \omega_m((\lambda y.N)(M \{N/x\}))$
 $\leq \max(\text{size}(A), \omega_m(M), \omega_m(N))$

$$(M, N) \neq (x u, \lambda y. v)$$

- $\omega_m(M M' \{N/x\}) = \omega_m((M \{N/x\})(M' \{N/x\}))$
 $\leq \max(\omega_m(M), \omega_m(M'), \omega_m(N), \text{size}(A))$
 $\leq \max(\dots)$

Exercise 8.

By induction on $\Gamma \vdash M : A$,

- case $\Gamma, x : A, \Delta \vdash x : A$, then $\omega_m(x) = 0$.

- case $\Gamma \vdash \lambda x. M : A \rightarrow B$, then $\Gamma, x : A \vdash M : B$,

$$\text{if } \omega_m(M) = 0 \text{ then } \omega_m(\lambda x. M) = 0$$

$$\text{if } \omega_m(M^*) < \omega_m(M) \text{ then}$$

$$\begin{aligned}
 \omega_m((\lambda x. M)^*) &= \omega_m(\lambda x. M^*) \\
 &= \omega_m(M^*) \\
 &< \omega_m(M) \\
 &= \omega_m(\lambda x. M)
 \end{aligned}$$

- case $\Gamma \vdash MN : A$ then $\Gamma \vdash M : B \rightarrow A$ and $\Gamma \vdash N : B$.

if $M = \exists x.L$, then $(MN)^* = L^* \{N^*/x\}$

and thus

$$\omega_m((MN)^*) = \omega_m(L^* \{N^*/x\})$$

$$\leq \max(\omega_m(L^*), \omega_m(N^*), \text{size}(A))$$

$$\omega_m(MN) = \omega_m(\exists x.L)N \leq \max(\omega_m(L), \omega_m(N), \text{size}(A))$$

and we conclude by ih on L and N .

otherwise $\omega(M^*N^*) \leq \max(\omega_m(M^*), \omega_m(N^*))$

$$\omega(MN) \leq \max(\omega_m(M), \omega_m(N))$$

and we conclude.

Tutorial n° 4

Exercise 1.

$$\text{dom } \beta = \{R / \exists P, R \beta P\}.$$

Let us show the following lemma:

Lemma: if F, F' are CBN-contexts and $R, P \in \text{dom } \beta \cup \text{dom } \mathcal{C} \cup \text{dom } cc$, such that $F[R] = F'[P]$ then $R = P$ and $F = F'$.

Proof by induction on F, F'

- if $F = F' = []$ then $F[R] = R = P = F'[P]$
- if $F[R] N = F'[P] M$ then we must have $N = M$ and $F[R] = F'[P]$, and we conclude by ih.
- if $F[R] = R = F[P] N$

then $R = (\lambda x. R') R''$ or $R = \mathcal{C} R'$ or $R = cc R'$
 and $P = (\lambda x. P') P''$ or $P = \mathcal{C} P'$ or $P = cc P'$
 if $F = []$, then impossible
 if $F = F''[] M$, then impossible.

For CBV, do it next time.

$\lambda x. \lambda y. x$ and $\lambda x. \lambda y. y$ are not $\xrightarrow[\ast]{\varepsilon cc \mathcal{C}}$.

Exercise 3.

$$Q1. M \left\{ \frac{[\alpha](LN)}{[\alpha]L} \right\} := \begin{cases} [\alpha](L \dots N) & \text{if } M = [\alpha]L \\ [\beta]L \dots \{ & \text{if } M = [\beta]L \text{ with } \beta \neq \alpha \\ \mu \alpha. L & \text{if } M = \mu \alpha. L \\ \mu \beta. L \dots \{ & \text{if } M = \mu \beta. L \text{ with } \beta \neq \alpha \\ \lambda x. L \dots \{ & \text{if } M = \lambda x. L \\ x & \text{if } M = x \\ L \dots \{ R \dots \{ & \text{if } M = LR \end{cases}$$

Q2. For λ -calculus, the substitution lemma is:
 if $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$ then $\Gamma \vdash M \{N/x\} : B$.

Now, if $\Gamma \vdash N : B \mid \Delta$ and $\Gamma \vdash M : C \mid \Delta, \alpha : B \Rightarrow A$ (*)

then $\Gamma \vdash M \left\{ \frac{[\alpha]LN}{[\alpha]L} \right\} : C \mid \Delta, \alpha : A$.

We can show it by induction on $(*)$, we only need to consider the yellow cases.

For the case $\mu \alpha. M$, then we must use rule μ for $(*)$ which is impossible as the premise with two α s, absurd!

For the $[\alpha] M$ case then

$$\frac{\Gamma \vdash ([\alpha] M) \{ \dots \} : C \quad | \quad \Delta, \alpha : A}{[\alpha] (M \{ \dots \}) N}$$

Exercise 68 from the notes.

By induction on A ,

- if $A = X$, then as $M' \sim_x N [\eta : \sigma \leftrightarrow \sigma']$
we have $(M', N) \in \eta(x)$
and by admissibility,

$$M \cong_{\sigma(x)} M' \quad \eta(x) \quad N \cong_{\sigma'(x)} N$$

implies $M \sim N [\eta : \sigma \leftrightarrow \sigma']$

- if $A = B \Rightarrow C$, then take $L \sim_B P [\eta : \sigma \leftrightarrow \sigma']$
and we need to prove $M'L \sim_c NP [\eta : \sigma \leftrightarrow \sigma']$.

By hypothesis and context closure, $\square L$

$$ML \cong_{\sigma(c)} M'L \sim_B NP [\eta : \sigma \leftrightarrow \sigma']$$

thus by induction hypothesis, we conclude

$$ML \sim_c NP [\eta : \sigma \leftrightarrow \sigma'].$$

- if $A = \forall X, B$, take C, C' and $R : c \leftrightarrow c'$,

we want to show that

$$M'C \sim_B N C' [\eta[x \mapsto R] : \sigma[x \mapsto c] \leftrightarrow \sigma'[x \mapsto c']]$$

by hypothesis, $M \cong_{\sigma(\forall X, B)} N$ thus, by context closure $([]c)$

$$MC \cong_{\sigma(B)[c/x]} M'C \sim_B N C' [\eta']$$

and we conclude by induction hypothesis.

Corollary: Given a closed type A , then

$$M \sim_A N \quad \text{iff} \quad M \cong_A N.$$

Exercise (Proof of Corollary 2.3.10):

$$\bullet M \cong_{A \Rightarrow B} N \quad \text{iff} \quad M \sim_{A \Rightarrow B} N$$

$$\text{iff } \forall U \sim_A V, \quad MU \sim_B NV$$

$$\text{iff } \forall U \cong_A V, \quad MU \cong_B NV$$

$$\bullet \text{ if } M \cong_{\forall X, B} N \quad \text{we have by context closure } MC \cong_{B\{C/X\}} NC.$$

for all C

$$\bullet \text{ if } \forall C, \quad MC \cong_{B\{C/X\}} NC, \quad \text{then } MC \sim_{B\{C/X\}} NC \quad \forall C,$$

$$\text{Let } R: D \hookrightarrow D', \text{ let us show } M \sim_B N [X \mapsto R],$$

by the first lemmas

$$MD \cong_{B\{D/X\}} ND \sim_B ND' [X \mapsto R]$$

which implies the result. It remains to show

$$\text{that} \quad ND \sim_B ND' [X \mapsto R]$$

$$\text{which is true as } N \sim_{\forall X, B} N \text{ implies } ND \sim_B ND' [X \mapsto R]$$

Exercise 70.

Suppose $\vdash M : \forall X, (X \Rightarrow X)$ for some closed term M .

We have to show that $M \cong_{\forall X, X \Rightarrow X \wedge X, \lambda x. x}$

$$\text{i.e. } \forall A, \quad MA \cong_{A \Rightarrow A} (\lambda x. \lambda x. x) A =_{\beta} \lambda a'. a'$$

$$\text{i.e. } \forall A a, \quad MAa \cong_A (\lambda a. a) a =_{\beta} a$$

i.e. $\forall A a, MA a \cong_A a$. Let A, a .

By parametricity, $M \sim_{\forall X, X \Rightarrow X} M$.

Define $\sigma(X) := A$ and $\eta(X) := \{(p, a) \mid p \cong_A a \cong_A a\}$.

Then, $MA \sim_{X \Rightarrow X} MA [\eta]$ and thus $MA a \sim_X MA a [\eta]$
(as $a \sim_X a [\eta]$)

which implies $MA a \eta(A) MA a$ which implies $MA a \cong_A a$.

Exercise. Consider $A := \forall X Y, X \Rightarrow (X \Rightarrow Y) \Rightarrow Y$.

Prove that if $M:A$ then $M \cong_A \lambda x y. \lambda x f f x$.

Let B, C , $u:B$ and $v:B \Rightarrow C$. Let us show

$$M B C u v \cong_C v u.$$

By parametricity, we have $M \sim_A M$, thus

$$\sigma := \begin{pmatrix} X \mapsto B \\ Y \mapsto C \end{pmatrix} \text{ and } \eta(X) := \{(p, q) \mid p \cong_B u \cong_B q\}.$$
$$\text{and } \eta(Y) := \{(p, q) \mid p \cong_C v u \cong_C q\}.$$

this implies

$$M B C u v \sim_Y M B C u v [\eta]$$

which implies $M B C u v \cong_C v u$, as

$$v \sim_{X \Rightarrow Y} v [\eta] \text{ as}$$

Exercise: Given M a closed term of type Bool , show

$$M \cong_{\text{Bool}} \text{False} \text{ or } M \cong_{\text{Bool}} \text{True}.$$

i.e. by ext, for all A , $\#$, $\#$, for all A' , $\#$, $\#$

$$M A \# \# \cong \# \text{ or } \#$$

By parametricity, $M \sim_{\text{Bool}} M$.

Define $\bar{c}(X) := A$ and $\eta(X) := \{(P, Q) \mid P \sim_A Q \text{ and } \begin{pmatrix} P \cong \# \\ \text{or} \\ P \cong \# \end{pmatrix}\}.$

Then, $M A \sim M A [\eta]$

then $M A \# \sim M A \# [\eta]$ as $\# \sim_x \# [\eta]$

then $M A \# \# \sim M A \# \# [\eta]$ as $\# \sim_x \# [\eta]$

$\langle a, b \rangle$

$\text{pair} := \lambda a b. \lambda x. \lambda f. f a b.$

(1) if $M : A \bar{\wedge} B$, then there exists $P : A$, $Q : B$,

$$M (A \bar{\wedge} B) \text{ pair} \cong \text{pair } P Q.$$

$$=_{\beta} \lambda x. \lambda f. f P Q$$

By parametricity,

$$M \sim_{A \bar{\wedge} B} M$$

then define $\phi(X) := A \bar{\wedge} B$

$$\eta(X) := \left\{ (k, k) \mid \exists P, Q, H \cong K \cong \langle P, Q \rangle \right\}$$

$$\text{then } M (A \bar{\wedge} B) \sim_{(A \Rightarrow B \Rightarrow X) \Rightarrow X} M (A \bar{\wedge} B)$$

$$= \bigcup_{\substack{P:A \\ Q:B}} [\langle P, Q \rangle]_{\sim}^2$$

$$M (A \bar{\wedge} B) \text{ pair} \sim_X M (A \bar{\wedge} B) \text{ pair}$$

and we have

$$\text{pair} \sim_{A \Rightarrow B \Rightarrow X} \text{pair}$$

$$\text{i.e. } \forall U \sim U', \forall V \sim V',$$

$$\text{pair } U V \sim \text{pair } U' V'$$

Exercise 2.

We will show that $M \cong \langle \pi_1 M, \pi_2 M \rangle$.

By parametricity, we have that

$$M \sim_{A_1 \bar{\wedge} A_2} M$$

and so, defining $\bar{\mathcal{O}}(X) := A_1 \bar{\wedge} A_2$ and $\eta(X) := \{(K, K') \mid K \cong K' \cong \langle \pi_1 K, \pi_2 K \rangle\}$,

we have that

$$M(A_1 \bar{\wedge} A_2) \sim_{\substack{A_1 \Rightarrow A_2 \Rightarrow X \\ \Rightarrow X}} M(A_1 \bar{\wedge} A_2) [\eta : \bar{\mathcal{O}} \hookrightarrow \bar{\mathcal{O}}]$$

$$\text{as } \text{pair}_{A_1, A_2} \sim_{A_1 \Rightarrow A_2 \Rightarrow X} \text{pair}_{A_1, A_2} [\eta : \bar{\mathcal{O}} \hookrightarrow \bar{\mathcal{O}}]$$

$$\text{i.e. } \forall P \sim_{A_1} P', \forall Q \sim_{A_2} Q', \text{pair } P Q \sim_X \text{pair } P' Q' [\eta : \bar{\mathcal{O}} \hookrightarrow \bar{\mathcal{O}}],$$

$$\langle P, Q \rangle \sim_X \langle P', Q' \rangle [\eta]$$

$$\text{i.e. } \forall P \cong_{A_1} P', \forall Q \cong_{A_2} Q', \langle P, Q \rangle \cong \langle P', Q' \rangle \cong \langle \pi_1 \langle P, Q \rangle, \pi_2 \langle P, Q \rangle \rangle.$$

by context closure

$\text{by } \beta$

Thus we have that

$$H := M(A_1 \bar{\wedge} A_2) \text{pair} \sim_X M(A_1 \bar{\wedge} A_2) \text{pair} [\eta : \bar{\mathcal{O}} \hookrightarrow \bar{\mathcal{O}}]$$

i.e.

$$H \cong H \cong \langle \pi_1 H, \pi_2 H \rangle.$$

It then suffices to show that $H \cong M$, i.e. by extensionality,

$$\forall B, \forall N : A_1 \Rightarrow A_2 \Rightarrow B, \quad H B N \cong_B M B N.$$

By parametricity, $M \sim_{A_1 \bar{\wedge} A_2} M$, and defining $\bar{\sigma}(X) := B$
 $\bar{\sigma}'(X) := A_1 \bar{\wedge} A_2$

we have $M \text{ B N } \sim_X M (A_1 \bar{\wedge} A_2) \text{ pair} = \eta \left[\eta : \bar{\sigma} \leftrightarrow \bar{\sigma}' \right]$ and $\eta(X) = \{(P, Q) \mid P \equiv Q \text{ B N}\}$.

which immediately concludes.

We have to show that $N \sim_{A_1 \Rightarrow A_2 \Rightarrow X} \text{pair}$, i.e. $\forall P \sim_{A_1} P', \forall Q \sim_{A_2} Q'$,

$$N \text{ P Q } \sim_X \text{pair P' Q'} [\eta : \bar{\sigma} \leftrightarrow \bar{\sigma}']$$

$$\text{i.e. } N \text{ P Q } \cong \langle P', Q' \rangle \text{ B N } =_{\beta} N \text{ P' Q'}$$

and we have that by context closure.

Exercise 1.

$$\boxed{\begin{array}{l} cc : (A \Rightarrow B) \Rightarrow A \Rightarrow A \\ \mathcal{C} : (A \Rightarrow B) \Rightarrow B \Rightarrow A \end{array}} \quad (\text{with } B = \perp)$$

$$A : \perp \Rightarrow A.$$

Define $cc' := \lambda f^{A \Rightarrow B \Rightarrow A} \mathcal{C}(\lambda g^{A \Rightarrow B} g(\lambda g))$.

$$\begin{aligned} E[cc' M] &\longrightarrow_{\beta} E[\mathcal{C}(\lambda g. g(M g))] \xrightarrow{\mathcal{C}^{CBN}} (\lambda g. g(M g))(\lambda \lambda y. E[\lambda y]) \\ &\longrightarrow_{\beta} (\lambda \lambda y. E[\lambda y])(\lambda (\lambda \lambda y. E[\lambda y])) \\ &\longrightarrow_{\beta} E[M (\lambda \lambda y. E[\lambda y])]. \end{aligned}$$

Define $\mathcal{C}' := \lambda f^{A \Rightarrow B \Rightarrow B} cc(\lambda g. A \lambda f g)$.

$$\begin{aligned} E[\mathcal{C}' M] &\longrightarrow E[cc(\lambda g. A \lambda f g)] \xrightarrow{*} E[A(M (\lambda \lambda y. E[\lambda y]))] \\ &\longrightarrow M(\lambda \lambda y. E[\lambda y]) \end{aligned}$$

Exercise 3.

Define case $A_1 \bar{\vee} A_2 : \forall X. A_1 \bar{\vee} A_2 \Rightarrow (A_1 \Rightarrow X) \Rightarrow (A_2 \Rightarrow X) \Rightarrow X$
 $:= \lambda X. \lambda p a b. p X a b.$

case $(A_1 \bar{\vee} A_2) (im_a U) im_1 im_2 \xrightarrow{\beta^*} (im_x U) (A_1 \bar{\vee} A_2) im_1 im_2$
 $\xrightarrow{\beta^*} im_x U.$

Exercise 4.

Suppose $\exists M : A_1 \exists N : A_2, im_1 M \cong im_2 N$

and thus by context closure

case Bool $(\lambda _ true) (\lambda _ false) (im_1 M) \cong$ case Bool $(\lambda _ true) (\lambda _ false) (im_2 N)$
 $\parallel_{\beta}$ $\parallel_{\beta}$
 $true \cong false$

which is absurd.

Exercise 5.

By parametricity, $M \sim_{A_1 \bar{\vee} A_2} M$

thus $M(A_1 \bar{\vee} A_2) \sim_{(A_1 \Rightarrow X) \Rightarrow (A_2 \Rightarrow X) \Rightarrow X} M(A_1 \bar{\vee} A_2) [\eta : \mathcal{C} \hookrightarrow \mathcal{C}]$

Define $\mathcal{C}(X) := A_1 \bar{\vee} A_2$ $\eta(X) := \{(P, Q) \mid \exists R, P \cong Q \cong im_1 R\} \cup \{(P, Q) \mid \exists R, P \cong Q \cong im_2 R\}$

$M(A_1 \bar{\vee} A_2) im_1 im_2 \sim_X M(A_1 \bar{\vee} A_2) im_1 im_2 [\eta : \mathcal{C} \hookrightarrow \mathcal{C}].$

which then implies that either $M(A_1 \bar{\vee} A_2) im_1 im_2 \cong im_1 R$ for some R
 α $M(A_1 \bar{\vee} A_2) im_1 im_2 \cong im_2 R$ for some R

We only have to show that

$im_i \sim_{A_1 \Rightarrow X} im_i [\eta : \mathcal{C} \hookrightarrow \mathcal{C}]$

i.e. $\forall U \sim_{A_i} V, im_i U \sim_X im_i V [\eta : \mathcal{C} \hookrightarrow \mathcal{C}].$

i.e. $\forall U \cong_{A_1} V, \text{ im}_1 U \cong \text{im}_2 V$ with $R = U$
 $\hookrightarrow$ by context-closure.

Then, it remains to show that

$$M \cong_{A_1 \bar{\vee} A_2} M(A_1 \bar{\vee} A_2) \text{ im}_1 \text{ im}_2$$

i.e. by ext,

$$\forall B, U: A_1 \Rightarrow B \quad M B U V \cong_B M(A_1 \bar{\vee} A_2) \text{ im}_1 \text{ im}_2 B U V.$$

$$V: A_2 \Rightarrow B$$

By parametricity,

$$M \sim_{A_1 \bar{\vee} A_2} M$$

and, defining $\bar{\sigma}(X) := B$ $\bar{\sigma}'(X) := A_1 \bar{\vee} A_2$

$$\text{and } \eta(X) := \{(P, Q) \mid P \cong Q B U V\}_r$$

we have

$$M B \sim_{(A_1 \Rightarrow X) \Rightarrow (A_2 \Rightarrow X) = \bar{\vee} X} M(A_1 \bar{\vee} A_2) \quad [\eta : \bar{\sigma} \leftrightarrow \bar{\sigma}']$$

$$\text{and } M B U V \sim_B M(A_1 \bar{\vee} A_2) \text{ im}_1 \text{ im}_2 \quad [\eta]$$

$$\text{as } U \sim_{A_1 \Rightarrow X}^{\text{im}_1} [\eta] \text{ i.e. } \forall R \cong_{A_1} S,$$

$$UR \cong (\text{im}_2 S) B U V$$

$$\parallel_B$$

$$U S$$

and thus $M B U V \cong_B M(A_1 \bar{\vee} A_2) \text{ im}_1 \text{ im}_2$ by context closure.